Global Facilities - Design & Construction Standard - BIM - Execution Plan Guideline

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1.0 Purpose

1.1 The intent of the BIM Execution Plan (BEP) is to define a foundational framework to ensure successful deployment of advanced design technologies on Micron BIM enabled project. The BEP is about optimizing work and model flow across the project. The key is good planning of the design-to-engineering-to-construction process to minimize downstream surprises, rework, redundancies or gaps in the flow of (model-based) information.

2.0 Scope

2.1 The BEP is to be developed and utilized on all new design projects ranging from Greenfield projects to renovation projects. Projects with multiple team members on both the design and build side, a BEP will be required of the AE and the GC to better synchronize the BIM workflow.

3.0 Safety

3.1 This document was evaluated for potential job hazards. This procedure includes no recognized risks and a Job Hazard Analysis. If changes are required to this procedure, those changes must be evaluated for potential job hazards.

4.0 Definitions and Acronyms

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
BIM	Building Information Modeling
BEP	BIM Execution Plan
DBB	Design Bid Build
DB	Design Build
AE	Architect and Engineering Designers
GC	General Contractor

5.0 Roles and Responsibilities

	Responsibilities
GFTT BIM Lead	• Develop and maintain guidelines. • Perform periodic audits to ensure standards set are maintained at execution. • Document and incorporate all lessons learned into future BEP templates and standards.
FCT BIM Lead (BIM Coordinator / Engineer)	• Ensure Site Construction team draft out the contents as required in the BEP template prior to project start. • Work with Site Construction team to work out areas that require customization and feedback to GFTT BIM Lead. • Perform QA/QC checks on the BIM deliverables and ensure execution as per BEP.
AE Team BIM Lead	• Draft contents of the Design stage BEP relevant to the project base on BEP template issued by Micron. • Execute BIM according to BEP.
GC Team BIM Lead	• Draft contents of the Construction stage BEP relevant to the project base on BEP template issued by Micron. • Execute BIM according to BEP.
Site Facilities Teams	• Verify at close-out the standards documented in the BEP are adhered to.

6.0 BIM Execution Plan Overview

6.1 As a minimum requirement, the BEP must complete, submit and have 1st complete version approval by Micron within 30 working days after the project kickoff meeting. The contents of the BEP are to be developed in a collaborative environment at the project's BIM kick-off meeting. The data entry to, and distribution of, the projects BEP is the responsibility of the respective AE/GC BIM Leads unless otherwise appointed. Any additional planning information should also be included.

6.2 The AE/GC BIM Leads shall schedule follow-up BIM meetings to review contents of the BEP with all stakeholders periodically. The BEP is a live document and will act as a reference document during and after BIM delivery for future BIM maintenance needs. Periodic meetings shall be held to update the contents throughout the project lifecycle.

6.3 In the case of a DBB project delivery, AE Team shall draft the BEP for design execution needs with the objective of providing coordinated design BIM deliverables sufficient for Construction Team to construct from. If construction partners are available in the design stage, the AE is strongly encouraged to discuss the design BEP with the construction partners to provide a more seamless transfer to the Construction Team at construction stage.

6.4 Upon taking over the design BIM deliverables, The Construction Team shall draft the BEP for construction to as-built needs with the objective of delivering information rich BIM models for Micron to effectively manage the built assets.

6.5 In the case of a DB project delivery, the appointed BIM Lead shall draft the BEP for the complete BIM delivery from design to as-built with the objective of providing coordinated design BIM deliverables sufficient for Construction Team to construct from and final delivery of information rich BIM models for Micron to effectively manage the built assets.

7.0 BEP Template File

7.1 The latest BEP template file [LINK] is a document for distribution and is to be completed by the appointed party, printed into a PDF format, and placed in a central location (agreed at the BIM kick-off meeting), where all project participants will have easy access to the completed document for use.

7.2 The BEP can be completed with the priority items/sections, agreed with FCT BIM Lead, within 2 weeks after the completion of the BIM kick-off meeting and the issuance of the BEP template. The items not identified as priority items may be completed thereafter.

8.0 BEP – What to Include

8.1 Basic Information Requirements - The BEP must capture the basic project facts. The BEP template includes sections to document the following basic information:

- BEP Plan Overview: State the purpose of the BEP and a brief on how it is to be used throughout the project. Outline the overall BIM execution strategy and goals. This section is an executive summary and should be written on a macro level to allow quick understanding on the executions of BIM for the project.
- Project Information: Include project reference information such as, Project Owner, Project Name, Location, Address, Description, Project Number.
- Project Schedule: Include project schedule, phases, milestones to ensure BIM are delivered at the appropriate timing.
- BIM Team Organization and Key Contacts: Identify the responsible people with organization charts, document the BIM roles and responsibilities, responsibility matrix(RACI / ARCI), list their contact details.

8.2 Define the BIM Deliverables and BIM Uses

- BIM deliverables for the projects are to be listed here, which includes deliverables for close-out and each milestone. A complete list of models and drawings and how they relate to each other shall be provided and updated at close-out.
- Modelling detail progression shall be indicated in the milestone delivery and detail definition shall be included in the appendix for referencing.
- BIM uses are to be listed and clearly defined to ensure they are in line with the BIM strategy and goals as defined in the earlier section of the BEP.

8.3 BIM Standards – In order to achieve consistency of the delivery and deliverables by all team members, a set of BIM standards shall be defined and documented as follows:

- Naming Conventions: Consistent naming used in the BIM project is the foundation to many uses from clash detection setup, quantity, takeoff, ease of retrieving elements etc. This section shall contain all naming conventions to be used. Names for and not limited to, model files, folders, components, system, etc are to be clearly stated here.
- File Storage Structure: Files are to be stored and kept in a structured manner for proper placement of data and information. Folder structure to be defined here and shall be determined before any data exchange is to start.

- Model Geometry and Data Development Requirements: The detail level of the BIM models must be managed in order not handicap the designing progress, analysis efforts, constructability studies, model navigation, etc. Data required for each stages the project execution are paramount to aiding decision making. LOD definitions, data requirements along with the appropriate timing shall be defined here. Ensure that appropriate detail levels are required at the appropriate timing in order not to hinder work progress.
- Model Drafting and Colour Standards: To ensure deliverables has consistency in presentation, drafting standards like, and not limited to, paper size, standard tags, line weights, units of measurement to be used, discipline drafting and system colours, shall be defined here. The standards outlined here are to derive base on producing drawings direct from within the 3D authoring tool.

8.4 BIM Model Setup, Collaboration and Workflows – BIM requires multi-disciplines to work together and collaborate to ensure creation of BIM models that works. Below are the important information that must be capture in the BEP for this section:

- Model Setup: Model common or shared positioning parameters, coordinates, elevations, orientation, reference point for tool installations, unit of measurement, etc. are to be agreed and documented at the early stage of each design process.
- Model Breakdown: As different disciplines will work in their respective models and in order to keep the size of the models files at a manageable level, model breakdown is inevitable and shall be clearly listed as the project progresses. As the level of details develop over the course of the project, further splitting of the BIM models will be required and therefore need to be updated to this list.
- Model and Drawing List: 2D drawings and 3D geometry should correlate with each other and it will be expected that the 2D drawings are to be developed within the 3D authoring software. As such, where the drawings reside in which models shall be accurately documented here.
- Collaboration and Clash Management: A detail summary on the strategies for collaboration and clash management. Collaboration parties are to be listed. Clash settings, clash management and collaboration workflows, clash monitoring and reporting shall be clearly stated. Space management and optimization strategies shall be included.
- Processes and Workflows: This section shall provide the workflows that are relevant for the project in swimlane flow charts that includes information on how to perform, who is responsible to perform, what are the deliverables at each stage and when the tasks are to be performed (where appropriate).

These workflows shall include, but not limited to:
1) Design
2) Pre-Construction
3) Design Change by AE & GC
4) Construction
5) As-Built
6) BIM Modelling
7) Clash Management
8) Data Exchange

8.5 Quality Control

- Quality control checks and procedures are to be in place in order to ensure the consistency and minimal disruptions during the BIM execution, and final deliverables can be delivered as expected. This section shall indicate 1) the strategy to adopt, 2) "Quality Control Checks" matrix to make sure the team knows exactly what are to be checked.

8.6 Technology Requirements, Data exchange and Meetings

- These sections of the BEP shall list the technologies like, software, tools, platforms, etc., to be used in the project execution. How data are to exchange between the teams using the data exchange tool or platform listed are to be indicated here.
- To ensure information, collaboration and progress for the project are effectively communicated between the team members, all types of meetings, meeting purpose, frequency, their expected start dates and completion dates shall be listed here.

8.7 Additional Information

- Since no BEP template can accommodate all project's requirements and information, the final section of the BEP Template is for adding any supporting or additional information that might not be able to be captured in the previous sections.

9.0 Document Control

9.1 Approval: GLOBAL_FAC_GFTT_APPR

9.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

9.3 Final Viewer Access

All Micron Team Members at all Sites

9.4 Retention

Adherence to the requirements specified in the
Global Facilities – Document Control Procedures

9.5 Review

Biennially (two years)

10.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	08/22/21